

Bonus Research Assignment

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Instructions

This bonus assignment is intended to encourage deeper thinking about the physical concepts covered in this course and to help you connect mathematical tools from vector analysis to real physical phenomena in electrostatics and magnetostatics. In this assignment, you will prepare a short research-style write-up (similar to a long abstract) on a topic related to the course material.

Your submission should clearly explain the selected concept and discuss its physical meaning, mathematical formulation, and practical relevance in engineering or science. The focus should be on understanding and explaining the idea clearly rather than providing

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The write-up should follow a short research-style format. This means the document should be typed (not handwritten) and structured clearly. Figures and tables may be included when they help illustrate the concept, and each figure or table should have a caption explaining what is being shown. If helpful, you may include simple MATLAB simulations or plots to demonstrate the behavior of a field, potential, force, or other quantity. When using information from textbooks, lecture notes, or other sources, proper referencing should be provided. The goal is not to produce a full research paper, but rather a concise technical explanation similar to a long abstract.

The length of the write-up should be approximately **1–2 pages (maximum 2 pages)**, excluding references if needed. Submissions must be typed and submitted as a single PDF file.

Suggested topics include:

- Why do gradient, divergence, and curl matter in real life?**
Explain how these vector operators help describe physical phenomena such as heat flow, fluid motion, or electromagnetic fields.
- Why do different coordinate systems exist?**
Discuss why Cartesian, cylindrical, and spherical coordinate systems are useful, and how choosing the right coordinate system simplifies solving electromagnetic problems.
- How symmetry simplifies physics: the power of Gauss’s law**
Explain how symmetry allows us to solve complex electric field problems easily, and give examples such as spherical, cylindrical, or planar charge distributions.
- Electric potential: why voltage is often easier to work with than electric field**
Discuss the relationship between electric field and potential, and why potential-based

methods are widely used in engineering.

5. **What happens to electric fields inside materials?**

Explain polarization, dielectric materials, and how materials influence electric field distribution.

6. **Laplace's and Poisson's equations: why do these equations appear everywhere in physics and engineering?**

Discuss why these equations are fundamental and where they are commonly applied.

7. **Capacitors in real life: from energy storage to supercapacitors**

Explain how capacitors store energy and discuss practical applications such as energy storage systems and electronic circuits.

8. **Magnetic fields produced by currents: comparing Biot–Savart law and Ampère's law**

Explain the physical meaning of these laws and when each method is easier to apply.

9. **Magnetic dipoles: why current loops behave like small magnets**

Explain magnetic moment and discuss examples such as electric motors, sensors, or magnetic storage devices.

10. **Inductors and magnetic energy: where is the energy actually stored?**

Discuss how magnetic fields store energy and why inductors resist changes in current.

11. **Magnetic materials: why some materials are strongly magnetic and others are not**

Explain diamagnetic, paramagnetic, and ferromagnetic materials, and why their behavior differs.

12. **Wireless charging and inductive coupling**

Explain how magnetic fields are used to transfer energy without wires.

13. **Electrostatic discharge (ESD): why small sparks can damage electronics**

Explain the physical mechanism behind electrostatic discharge and its importance in electronic design.

14. **Magnetic levitation: how magnetic forces can overcome gravity**

Explain the physics behind magnetic levitation and examples such as maglev trains.

15. **How electromagnetic concepts are used in modern technologies**

Discuss applications such as sensors, communication systems, medical imaging, or microelectronic devices.

You may also choose another topic related to the course, provided that it is sufficiently connected to the material covered in lectures.

This is a bonus assignment worth **10%**, which will be applied toward compensating for marks that may have been lost in quizzes, midterms, or assignments.

Please submit your work as a single PDF file on Avenue to Learn.

Due on Apr 20, 2026 11:59 PM

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